THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 10 — SCIENCE · HARD LEVEL CHAPTER TEST SET 1

Chapter 10: **The Human Eye and the Colourful World** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Assertion (A): The image formed on the retina is inverted. Reason (R): The brain interprets the signals to make the object appear erect. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

2. A person with a far point at 25 cm requires spectacles to see distant objects clearly. The focal length of the lens is 1

(a) -6 m **(b)** +25 cm **(c)** -25 cm **(d)** -31 cm

3. The sky is blue because 1

(a) larger wavelengths are scattered more by small particles
(b) smaller wavelengths are scattered more by small particles
(c) the sun emits blue only
(d) the sea reflects blue

4. Assertion (A): A myopic person cannot see distant objects clearly. Reason (R): In myopia the image of a distant object is formed behind the retina. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

5. A person can see only from 40 cm to 100 cm. The lens power for distance and reading lens power (to read at 25 cm) are respectively 1

(a) -0.4 D and +3 D **(b)** -1 D and +1.5 D
(c) -2.5 D and +1.5 D **(d)** -1 D and +1 D

6. The sky appears blue because molecules of the atmosphere scatter 1

(a) blue light more strongly than red light **(b)** all colours equally
(c) red light more strongly than blue light **(d)** only yellow light

7. Assertion (A): Violet light is deviated most by a prism. Reason (R): The refractive index of glass is maximum for violet light. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

8. The ability of the eye to see objects clearly at different distances depends on 1

(a) the cornea changing its colour
(b) the ciliary muscles changing the shape of the lens
(c) the retina moving back and forth
(d) the pupil moving the lens

9. The size of the particles in the atmosphere responsible for the white colour of clouds is 1

(a) zero **(b)** about the size of an atom

(c) larger than the wavelength of light **(d)** much smaller than the wavelength

10. Assertion (A): The sky appears blue. Reason (R): Blue light is scattered the most by air molecules. 1

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Why does the amount of scattering of red light not matter much in the colour of the sky? 2

12. If the angle of the incidence on the prism is varied, will the colours separate in the same way? State why. 2

13. How does the iris protect the eye and how does the lens focus light? 2

14. Calculate the power of the lens required by a person whose near point is 50 cm to read at 25 cm. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A person with a near point at 30 cm wants to read at 25 cm. Find the power of the lens required. 3

16. Why does the sky appear whitish near the horizon on a clear day? 3

17. An object is placed at 10 cm and 100 cm in front of a normal eye. At which position does the ciliary muscle contract more and why? 3

OR

An elderly person has a near point of 1 m and a far point of 2 m. Describe the defects and the lens prescription for reading at 25 cm and seeing far. 3

18. If a person looks at the Sun through a prism in the evening, why does he see colours at the edge of the disc? 3

OR

Why can't we see the stars in the daytime while they can be seen at night? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A person with a far point at 90 cm and a near point at 25 cm is prescribed a lens. 5

(a) Find its power and focal length

(b) state the nature

(c) find the new far point if a lens of power -1 D is used instead of the correct one.

OR

An eye of a person has near point at 25 cm and far point at 2 m. Find the power of lens that can bring distant objects into clear focus. Predict the nature of the image formed by the lens of an object at infinity. 5

20. A hypermetropic person has a near point of 0.5 m. 5

(a) Find the power of the corrective lens to read at 25 cm.

(b) Find the new near point with a lens of +1 D.

OR

Explain how the knowledge of scattering of light is used in 5

(a) danger signals

(b) fog lamps

(c) sky colour on earth vs. Moon. Give reasons.

END OF QUESTION PAPER — Total: 40 Marks